PRAKTIKUM BASIS DATA

MODUL 8

Scripting SQL Create

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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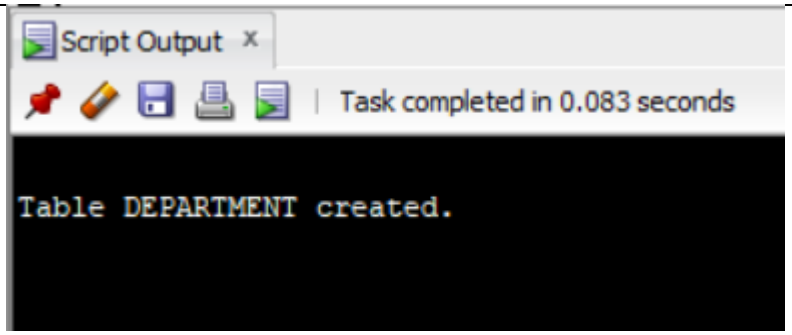
Jurnal

1. Buatlah desain database untuk suatu perusahaan:
   a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.

      Query

```
CREATE TABLE department (
department_id   NUMBER(4) PRIMARY KEY,
department_name VARCHAR2(100),
manager_id      NUMBER(6),
location_id     NUMBER(4)
);
```

      Screenshot

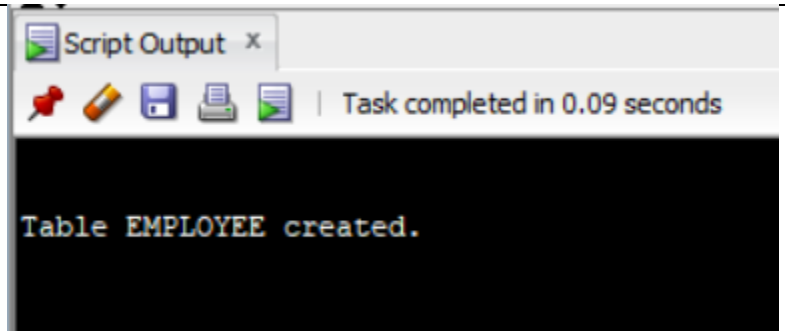


   b. Table employee dengan constraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_name NOT NULL, email, salary, commission_pct, hire_date NOT NULL.

Query

```
CREATE TABLE employee (
employee_id       NUMBER(6) PRIMARY KEY,
last_name         VARCHAR2(100) NOT NULL,
email             VARCHAR2(100),
salary            NUMBER(10,2),
commission_pct    NUMBER(4,2),
hire_date         DATE NOT NULL,
department_id     NUMBER(4),
CONSTRAINT emp_dept_fk FOREIGN KEY (department_id)
REFERENCES department(department_id)
);
```

Screenshot



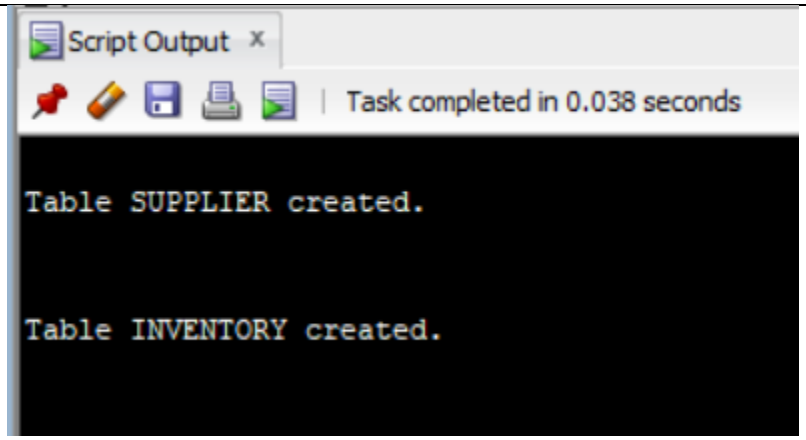
c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).

Query

```
CREATE TABLE supplier (
supplier_id     NUMBER(4) PRIMARY KEY,
supplier_name   VARCHAR2(100),
contact_name    VARCHAR2(100),
phone_number    VARCHAR2(20)
);

CREATE TABLE inventory (
inventory_id    NUMBER(4) PRIMARY KEY,
item_name       VARCHAR2(100),
quantity        NUMBER(6),
price           NUMBER(10,2),
supplier_id     NUMBER(4),
CONSTRAINT inv_supplier_fk FOREIGN KEY (supplier_id)
REFERENCES supplier(supplier_id)
);
```

Screenshot

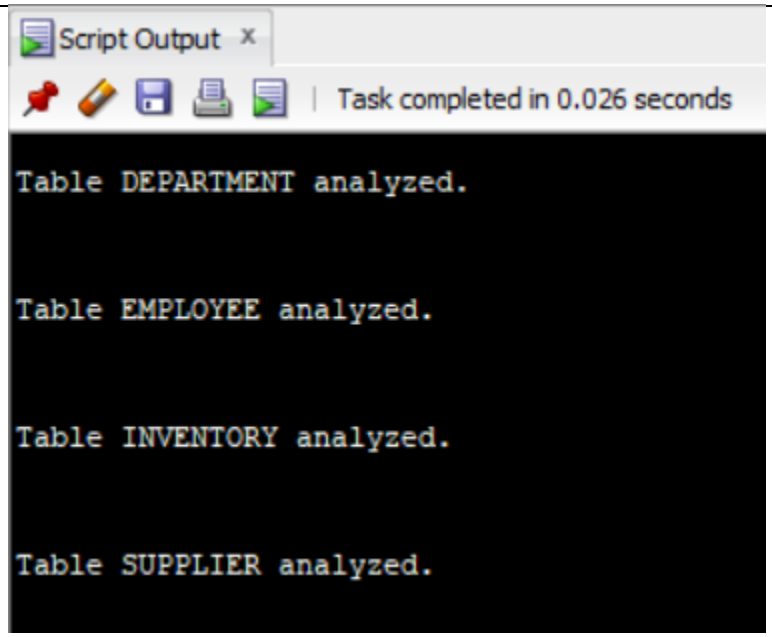


2. Analisis table-table dengan compute statistics

Query

```
ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE inventory COMPUTE STATISTICS;
ANALYZE TABLE supplier COMPUTE STATISTICS;
```

Screenshot



3. Isikan data-data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing-masing 10 record).

Query

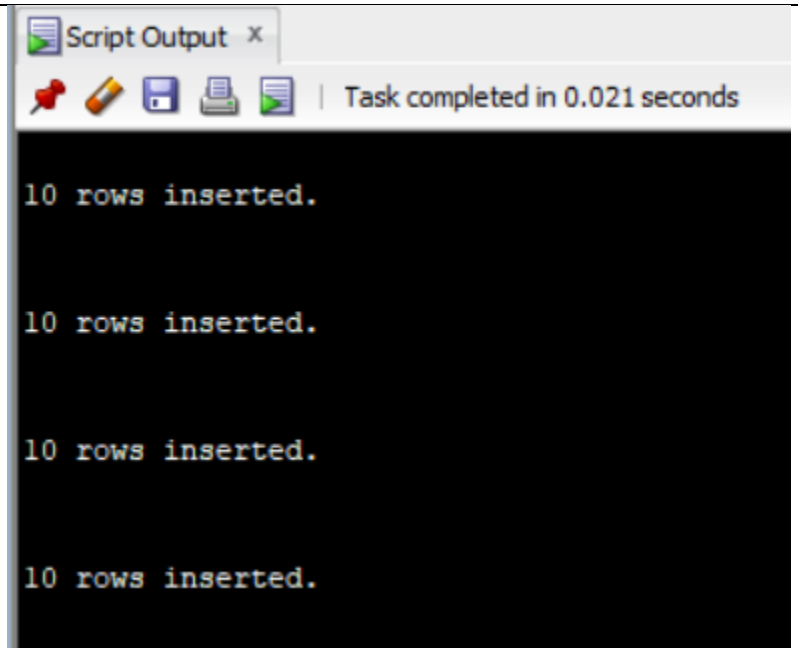
```
INSERT ALL
    INTO department VALUES (10, 'HR',              101, 1)
    INTO department VALUES (20, 'Finance',         102, 2)
    INTO department VALUES (30, 'IT',              103, 3)
    INTO department VALUES (40, 'Sales',           104, 4)
    INTO department VALUES (50, 'Marketing',       105, 5)
    INTO department VALUES (60, 'R and D',         106, 6)
    INTO department VALUES (70, 'Legal',           107, 7)
    INTO department VALUES (80, 'Customer Service', 108, 8)
    INTO department VALUES (90, 'Operations',      109, 9)
    INTO department VALUES (100, 'Logistics',      110, 10)
SELECT * FROM dual;

INSERT ALL
    INTO employee VALUES (1,  'Smith',    'john.smith@example.com',    50000, 0.10, DATE '2020-01-01', 10)
    INTO employee VALUES (2,  'Johnson',  'jane.johnson@example.com',   60000, 0.15, DATE '2019-02-01', 20)
    INTO employee VALUES (3,  'Williams', 'mark.williams@example.com',  70000, 0.20, DATE '2018-03-01', 30)
    INTO employee VALUES (4,  'Brown',    'lisa.brown@example.com',     80000, 0.25, DATE '2017-04-01', 40)
    INTO employee VALUES (5,  'Davis',    'paul.davis@example.com',     55000, 0.18, DATE '2021-05-01', 50)
    INTO employee VALUES (6,  'Miller',   'anna.miller@example.com',    65000, 0.22, DATE '2020-06-01', 60)
    INTO employee VALUES (7,  'Wilson',   'susan.wilson@example.com',   75000, 0.30, DATE '2019-07-01', 70)
    INTO employee VALUES (8,  'Moore',    'david.moore@example.com',    85000, 0.35, DATE '2018-08-01', 80)
    INTO employee VALUES (9,  'Taylor',   'luke.taylor@example.com',    95000, 0.40, DATE '2017-09-01', 90)
    INTO employee VALUES (10, 'Anderson', 'rachel.anderson@example.com',45000, 0.05, DATE '2022-10-01', 100)
SELECT * FROM dual;
```

```
INSERT ALL
    INTO supplier VALUES (1, 'Tech Supplier Inc.',    'John Doe',   '1234567890')
    INTO supplier VALUES (2, 'Office Supplies Ltd.',  'Jane Smith', '9876543210')
    INTO supplier VALUES (3, 'Gadget Store',          'Mark Brown', '5555555555')
    INTO supplier VALUES (4, 'Electronic Goods Co.',  'Sarah White','4444444444')
    INTO supplier VALUES (5, 'Print Solutions',       'Paul Green', '3333333333')
    INTO supplier VALUES (6, 'Digital Source',        'Kevin Lee',  '2222222222')
    INTO supplier VALUES (7, 'Smart Devices',         'Lisa Ray',   '1111111111')
    INTO supplier VALUES (8, 'Computer Hub',          'Andi Putra', '6666666666')
    INTO supplier VALUES (9, 'Office Mart',           'Sinta Dewi', '7777777777')
    INTO supplier VALUES (10,'Global Elektronik',     'Rudi Harto', '8888888888')
SELECT * FROM dual;

INSERT ALL
    INTO inventory VALUES (1,  'Laptop',     20, 10000000, 1)
    INTO inventory VALUES (2,  'Mouse',      50,    150000, 2)
    INTO inventory VALUES (3,  'Keyboard',   30,    250000, 3)
    INTO inventory VALUES (4,  'Monitor',    10,   2000000, 4)
    INTO inventory VALUES (5,  'Printer',    15,   1500000, 5)
    INTO inventory VALUES (6,  'Hard Drive', 40,    750000, 6)
    INTO inventory VALUES (7,  'Webcam',     25,    500000, 7)
    INTO inventory VALUES (8,  'Speakers',   35,    600000, 8)
    INTO inventory VALUES (9,  'USB Cable', 100,     50000, 9)
    INTO inventory VALUES (10, 'Phone',      10,   3000000, 10)
SELECT * FROM dual;
```

Screenshot



4. Analisis table-table dengan compute statistics

Query

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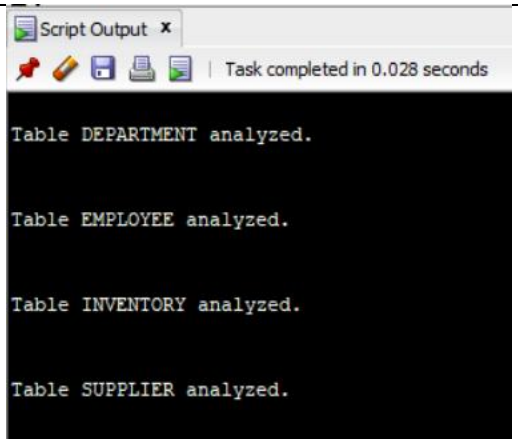


5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.

   Query

```
CREATE OR REPLACE VIEW empvu80 AS
SELECT
employee_id AS id_number,
last_name   AS name,
salary,
department_id
FROM employee
WHERE department_id = 80;
```

   Screenshot

```
Script Output
Task completed in 0.265 seconds

View EMPVU80 created.
```

6. Kemudian tampilkan struktur dari view empvu80.

   Query

```
DESC empvu80;
```

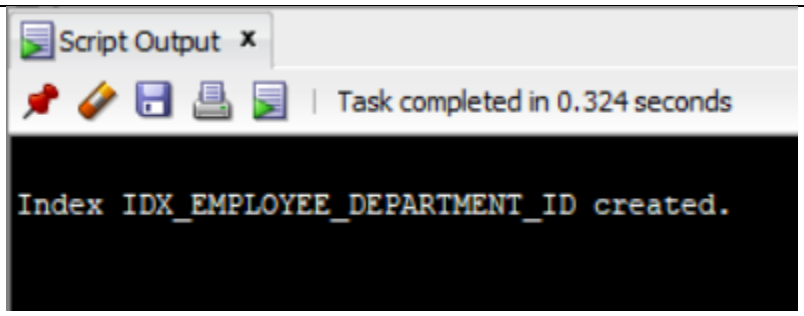
   Screenshot

```
Script Output
Task completed in 0.023 seconds

Name          Null?    Type
------------- -------- -------------
ID_NUMBER     NOT NULL NUMBER(6)
NAME          NOT NULL VARCHAR2(100)
SALARY                 NUMBER(10,2)
DEPARTMENT_ID          NUMBER(4)
```

7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE.

   Query

```
CREATE INDEX idx_employee_department_id
ON employee(department_id);
```

   Screenshot